

Memory

Overview

Memory is the second major component of a computer system. Ideally, memory should be:

- **Very fast** so the CPU does not wait,
- **Very large** so it can store plenty of data,
- **Very cheap** so the system remains affordable.

No current technology can satisfy all three goals at the same time. Therefore, computer systems use a **memory hierarchy**, where different layers provide different trade-offs between **speed**, **capacity**, and **cost per bit**.

1. Memory Hierarchy

Main Idea

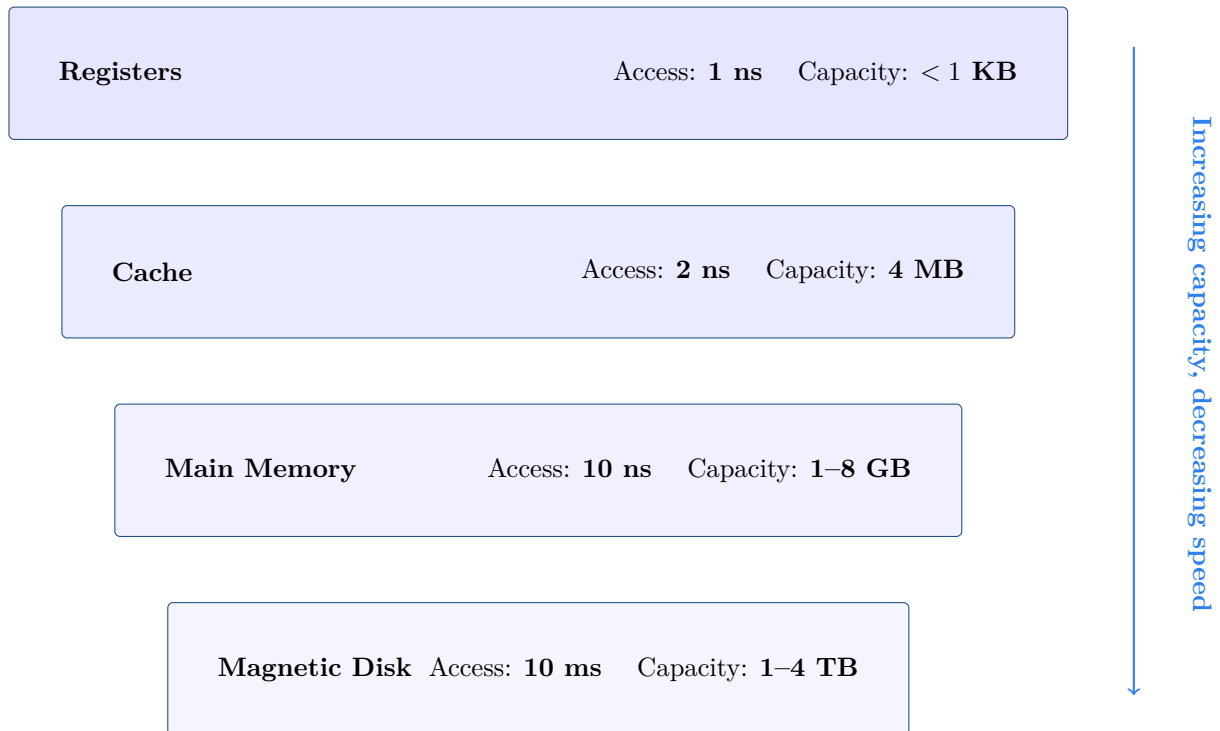
The memory system is arranged in layers. The upper layers are:

- faster,
- smaller in size,
- more expensive per bit.

The lower layers are:

- slower,
- larger in size,
- cheaper per bit.

1.1 Typical Memory Hierarchy Diagram



Observation

As we move downward in the memory hierarchy:

Speed decreases but Capacity increases

This layered design balances overall system performance and cost.

2. Registers

Definition

Registers are the top layer of the memory hierarchy. They are located inside the CPU and are made of the same material as the processor itself. Therefore, they are extremely fast and can be accessed without delay.

Key Features of Registers

- Registers are the **fastest** form of memory.
- Their capacity is **very small**.
- A 32-bit CPU may have around **32 registers of 32 bits each**.
- A 64-bit CPU may have around **64 registers of 64 bits each**.
- Total storage is usually **less than 1 KB**.
- Programs must manage registers in software, deciding which values should remain there.

3. Cache Memory

Definition

Cache memory is a small, high-speed memory placed inside or very close to the CPU. It stores the most frequently used portions of main memory so that the CPU can access data more quickly.

3.1 How Cache Works

Basic Working Principle

Main memory is divided into fixed-size blocks called **cache lines**, usually **64 bytes** each. For example:

- Cache line 0 stores addresses 0–63
- Cache line 1 stores addresses 64–127
- Cache line 2 stores addresses 128–191

When the CPU needs a word from memory, the cache hardware first checks whether the required line is already present in the cache.

3.2 Cache Hit and Cache Miss

Important Terms

- **Cache Hit:** The required data is found in the cache. The request is served immediately.
- **Cache Miss:** The required data is not found in the cache. The system must fetch it from main memory, which is slower.

Performance Note

Cache hits are very fast and may take only about **two clock cycles**. Cache misses are much slower because the system has to access main memory.

3.3 Why Cache is Useful

Caching Beyond RAM

Caching is widely used whenever some items are used more often than others. Examples include:

- Keeping heavily used file data in main memory,
- Storing results of pathname lookups,
- Storing URL-to-IP address conversions,
- Improving operating system performance in many repeated tasks.

3.4 Important Cache Design Questions

Common Questions in Any Caching System

In a caching system, several questions arise:

1. When should a new item be placed into the cache?
2. Which cache line should hold the new item?
3. Which old item should be removed if space is needed?
4. If an item is evicted, where should it be written back in larger memory?

3.5 Address Mapping Example

Example

Suppose:

- there are **4096 cache lines**,
- each line holds **64 bytes**,
- memory addresses are **32 bits**.

Then:

- bits **0–5** may specify the byte inside the cache line,
- bits **6–17** may specify the cache line number.

3.6 Levels of Cache

L1 and L2 Cache

Modern CPUs commonly use multiple levels of cache:

- **L1 Cache:** Smallest and fastest, usually inside the CPU.
- **L2 Cache:** Larger but slightly slower than L1.

Details

- Most CPUs have one L1 cache for instructions and another L1 cache for data.
- Each L1 cache is typically around **16 KB**.
- L2 cache may hold **several megabytes** of recently used memory words.
- L1 cache access is almost immediate.
- L2 cache access may involve a delay of **one or two clock cycles**.

3.7 Cache in Multicore Processors

Shared vs Separate L2 Cache

In multicore chips, designers must decide how the cache is organized:

- **Shared L2 Cache:** One L2 cache is used by all cores.
- **Separate L2 Cache:** Each core has its own L2 cache.

Both approaches have advantages and disadvantages:

- Shared L2 cache needs a more complex controller.
- Separate L2 caches make consistency management more difficult.

4. Main Memory

Definition

Main memory, usually called **RAM (Random Access Memory)**, is the main working memory of the computer. It stores programs and data currently in use by the CPU.

Features of Main Memory

- Main memory is larger than cache but slower.
- It is the **workhorse** of the memory system.
- All CPU requests not satisfied by cache go to main memory.
- Typical size ranges from **hundreds of megabytes** to **several gigabytes**.

Historical Note

Older computers used tiny magnetizable ferrite cores, so main memory was once called **core memory**. The old name still survives in informal usage.

5. Nonvolatile Memory

Meaning

Nonvolatile memory is memory that **does not lose its contents when power is turned off**.

5.1 ROM

ROM: Read Only Memory

- ROM is programmed at the factory.
- Its contents cannot normally be changed afterward.
- It is fast and inexpensive.
- It is often used to store the **bootstrap loader**.
- Some I/O cards also contain ROM for low-level device control.

5.2 EEPROM and Flash Memory

EEPROM and Flash

- **EEPROM** means Electrically Erasable PROM.
- Both EEPROM and flash memory are nonvolatile.
- Unlike ROM, they can be erased and rewritten.
- Writing them is much slower than writing to RAM.
- They are useful because stored programs can be corrected or updated later.

Flash Memory Uses

Flash memory is commonly used in portable electronic devices such as:

- digital cameras,
- portable music players,
- other handheld storage devices.

It is faster than disk but slower than RAM.

Limitation of Flash

Flash memory has a limited number of erase/write cycles. If it is erased too many times, it wears out.

6. CMOS Memory

Definition

CMOS memory is a small, low-power memory used in many computers to store system configuration and time-related information.

Uses of CMOS

CMOS memory is commonly used to store:

- current date and time,
- clock information,
- boot configuration,
- hardware setup parameters.

Why CMOS is Used

CMOS consumes very little power, so it can be supported by a small battery for several years, even when the computer is unplugged.

Battery Failure Effect

When the CMOS battery starts failing, the computer may “forget” important settings such as:

- current time and date,
- which hard disk to boot from,
- stored configuration parameters.

7. Summary Table

Memory Type	Main Characteristics	Volatile / Nonvolatile
Registers	Fastest memory, inside CPU, very small capacity	Volatile
Cache	Very fast, stores frequently used data, improves CPU speed	Volatile
Main Memory (RAM)	Main working memory, larger than cache, stores active programs and data	Volatile
ROM	Factory programmed, cannot usually be changed, used for boot code	Nonvolatile
EEPROM	Electrically erasable and rewritable, slower to write than RAM	Nonvolatile
Flash Memory	Rewritable, portable storage, limited write cycles	Nonvolatile
CMOS	Stores time and configuration using very low battery power	Volatile (battery-backed)

8. Conclusion

Final Note

The memory system of a computer is designed as a hierarchy because no single type of memory can provide maximum **speed**, **capacity**, and **low cost** together. Registers, cache, main memory, ROM, flash memory, and CMOS each serve a unique role. Together, they make the computer system efficient, practical, and cost-effective.